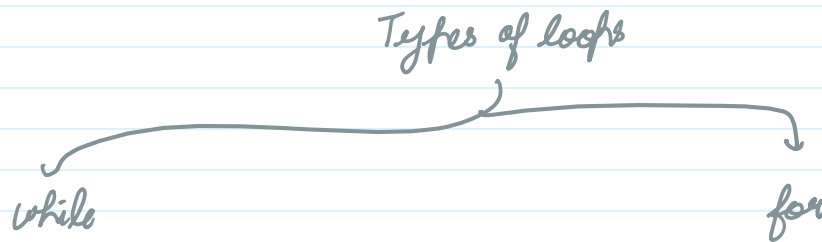


Nested Loops:-

→ Loop inside another loop.



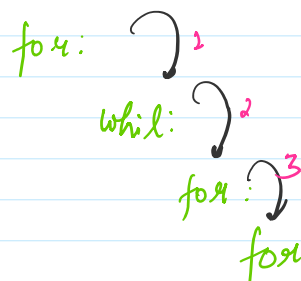
1) while :
while :

3) for :
for :

2) while :
for :

4) for :
while

Depth of nesting:- 1



→ depth of nesting = 3

Patterns

1)

```

x
x x
x x x
x x x x
x x x x x
  
```

Step 1:- Outer loop handle no. of rows.

Step 2:- Try to use nested inner loop and try connect/relate it with outer

```

* * *
* * * *
* * * * *

```

try connect/relate it with outer loop for printing stars/pattern.

Step 3: Relate outer & inner loop.

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	*				
R ₂	*	*			
R ₃	*	*	*		
R ₄	*	*	*	*	
R ₅	*	*	*	*	*

Observation:

for row in range(1,6): ← no. of rows

for col in range(row):

print("*")

print("\n")

Row no. = No. of stars

```

for row in range(1,6):
    for col in range(row):
        print("*",end="")

    print("\n")

```

```

for row in range(1,6):
    for col in range(row):
        print("*", end="")
    print("\n")

```

DRY RUN

Outer Loop (row)	Inner Loop (col)	Action	Output on screen
row = 1	col = 0	print("*", end="")	*
	(inner loop ends)	print("\n")	*\n
row = 2	col = 0	print("*", end="")	*
	col = 1	print("*", end="")	**
	(inner loop ends)	print("\n")	**\n
row = 3	col = 0	print("*", end="")	*
	col = 1	print("*", end="")	**
	col = 2	print("*", end="")	***
	(inner loop ends)	print("\n")	***\n
row = 4	col = 0	print("*", end="")	*
	col = 1	print("*", end="")	**
	col = 2	print("*", end="")	***
	col = 3	print("*", end="")	****
	(inner loop ends)	print("\n")	****\n
row = 5	col = 0	print("*", end="")	*
	col = 1	print("*", end="")	**
	col = 2	print("*", end="")	***
	col = 3	print("*", end="")	****
	col = 4	print("*", end="")	*****
	(inner loop ends)	print("\n")	*****\n

PATTERN OUTPUT

```

After row = 1
*

After row = 2
**

After row = 3
***

After row = 4
****

After row = 5
*****

```

0

2)

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	*	*	*	*	*
R ₂	*	*	*	*	*
R ₃	*	*	*	*	*
R ₄	*	*	*	*	*
R ₅	*	*	*	*	*

Observation:

5
5
5
5
5

for row in range(1,6):

for col in range(5):

print("*",end=" ")

\n